

Revision of the Grade 7 course

Powers, brackets, factorising and linear equations — the four tools every fraction question in this chapter will need.

LESSON 1–3

3 × 40 MIN

ALGEBRA 8, P. 3

STAGE 8 REVIEW

LESSON CLOCK

5'

12'

6'

15'

2'

Warm-up	5 min
Explanation	12 min
Interactive	6 min
Practice	15 min
Homework	2 min

LEARNING OBJECTIVES

- Apply the laws of indices to simplify products and quotients of powers.
- Expand brackets, including the product of two binomials.
- Factorise using a common factor, grouping, and the three standard identities.
- Solve a linear equation and rearrange a simple formula.

TEXTBOOK

National	pp. 3–6
Cambridge	Stage 9, Unit 2 (pre-work)
Workbook	—

01

Explanation

Why we start here

Everything in Chapter I — cancelling fractions, common denominators, adding and multiplying them — is really *factorising in disguise*. A fraction such as $\frac{x^2 - 9}{x + 3}$ only

simplifies once you see $x^2 - 9$ as $(x - 3)(x + 3)$. So three lessons on the Grade 7 toolkit is not lost time; it is the whole quarter's foundation.

1 · The laws of indices

For any $a \neq 0$ and whole numbers m, n :

$a^m \cdot a^n = a^{m+n} \quad \cdot \quad a^m \div a^n = a^{m-n} \quad \cdot \quad (a^m)^n = a^{mn} \quad \cdot \quad (ab)^n = a^n b^n$

Multiply powers of the same base — **add** the indices. Divide — **subtract**. A power of a power — **multiply**. Nothing else is allowed: $a^3 + a^3$ is $2a^3$, never a^6 .

THE CLASSIC SLIP

$(2a)^3 = 8a^3$, not $2a^3$ — the index reaches the 2 as well.

2 · Expanding brackets

Every term inside meets every term outside:

$$a(b + c) = ab + ac \quad \cdot \quad (a + b)(c + d) = ac + ad + bc + bd$$

Three products are worth knowing by heart, because you will need to read them *backwards* all quarter:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a - b)(a + b) = a^2 - b^2$$

3 · Factorising — the four moves

1. **Common factor.** $6x^2 - 9x = 3x(2x - 3)$

2. **Difference of two squares.** $4x^2 - 25 = (2x - 5)(2x + 5)$

3. **Perfect square.** $x^2 - 10x + 25 = (x - 5)^2$

4. **Grouping.** $ax + ay + bx + by = a(x + y) + b(x + y) = (a + b)(x + y)$

Always try the common factor first. It is the move learners forget, and it makes the other three easier.

4 · Linear equations and formulae

Clear the fractions, open the brackets, collect the unknowns on one side, divide.

Rearranging a formula is the same process with letters instead of numbers: to make

h the subject of $S = 2\pi rh$, divide both sides by $2\pi r$ to get $h = \frac{S}{2\pi r}$.

02 Worked examples**EXAMPLE 1** Simplify $(3x^2y)^2 \cdot 2xy^3 \div 6x^3y^4$

① $(3x^2y)^2 = 9x^4y^2$

The index 2 reaches the 3, the x^2 and the y .

② $9x^4y^2 \cdot 2xy^3 = 18x^5y^5$

Multiply the numbers, add the indices.

③ $18x^5y^5 \div 6x^3y^4 = 3x^2y$

Divide the numbers, subtract the indices.

ANSWER

$$3x^2y$$

EXAMPLE 2 *Factorise $2x^2 + 7x + 3$*

① $2 \cdot 3 = 6$

Multiply the first and last coefficients.

② $6 = 1 \cdot 6$, and $1 + 6 = 7$

Find two numbers with product 6 and sum 7.

③ $2x^2 + x + 6x + 3$

Split the middle term using those numbers.

④ $x(2x + 1) + 3(2x + 1)$

Group in pairs and take out the common factor of each pair.

⑤ $(2x + 1)(x + 3)$

The bracket $(2x + 1)$ is now itself a common factor.

ANSWER

$(2x + 1)(x + 3)$

EXAMPLE 3 *Solve* $\frac{x-2}{3} = \frac{x+4}{5}$

① $5(x-2) = 3(x+4)$
Cross-multiply — legal because 3 and 5 are never zero.

② $5x - 10 = 3x + 12$
Open both brackets.

③ $2x = 22$
Collect x on the left, numbers on the right.

④ $x = 11$
Divide by 2.

ANSWER

$x = 11$

03 Interactive model

Run through one factorisation together, then hand the mouse to a learner for the next.

Factorising, one step at a time

$$x^2 - 10x + 25$$

① x^2

is $(x)^2$

, 25

and

is 5^2

Look
for
two
perfect
squares.

② Check the middle term:

$2 \cdot x$ ✓

$\cdot 5$

=

$10x$

It
matches,
and
the
sign
is
minus.

3 $(x - 5)^2$

So
this
is a
perfect
square,
not
a
difference
of
squares.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Power	Daraja	Степень
Base of a power	Daraja asosi	Основание степени
Index (exponent)	Daraja ko'rsatkichi	Показатель степени
Expand the brackets	Qavslarni ochish	Раскрыть скобки
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Common factor	Umumiy ko'paytuvchi	Общий множитель
Difference of two squares	Kvadratlar ayirmasi	Разность квадратов
Perfect square	To'la kvadrat	Полный квадрат
Linear equation	Chiziqli tenglama	Линейное уравнение
Subject of a formula	Formuladan izlanayotgan kattalik	Выражаемая величина

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $a^5 \cdot a^3$ equals:

a^8

a^{15}

$2a^8$

a^2

 $(3x)^2$ equals:

$3x^2$

$9x^2$

$6x^2$

$9x$

Which is a difference of two squares?

$x^2 + 9$

$x^2 - 9$

$x^2 - 9x$

$x^2 + 6x + 9$

$x^2 - 6x + 9$ factorises to:

$$(x - 3)(x + 3)$$

$$(x - 9)(x + 1)$$

$$(x - 3)^2$$

$$x(x - 6) + 9$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify: $x^5 \cdot x^3$

ANSWER x^8

2. Simplify: $(a^3)^4$

ANSWER a^{12}

3. Expand: $3(2x - 5)$

ANSWER $6x - 15$

4. Expand: $(x + 4)(x - 4)$

ANSWER $x^2 - 16$

5. Factorise: $6a + 9$

ANSWER $3(2a + 3)$

6. Find the value of $2x^2 - 3x$ when $x = -2$

ANSWER 14

7. Solve: $5x - 7 = 3x + 9$

ANSWER $x = 8$

Medium · the standard question

7 PROBLEMS

1. Simplify: $(2a^2b)^3$

ANSWER $8a^6b^3$

2. Expand and simplify: $(x + 3)^2 - (x - 3)^2$

ANSWER $12x$

3. Factorise: $x^2 - 10x + 25$

ANSWER $(x - 5)^2$

4. Factorise: $4x^2 - 9y^2$

ANSWER $(2x - 3y)(2x + 3y)$

5. Factorise: $ax + ay + bx + by$

ANSWER $(a + b)(x + y)$

6. Solve: $\frac{x - 2}{3} = \frac{x + 4}{5}$

ANSWER $x = 11$

7. Simplify: $a^7 \div a^3 \cdot a^2$

ANSWER a^6

Hard · stretch and reason

7 PROBLEMS

1. Factorise completely: $x^3 - x$

ANSWER $x(x - 1)(x + 1)$

2. Factorise: $2x^2 + 7x + 3$

ANSWER $(2x + 1)(x + 3)$

3. Simplify: $(3x^2y)^2 \cdot 2xy^3 \div 6x^3y^4$

ANSWER $3x^2y$

4. Factorise completely: $x^4 - 16$

ANSWER $(x - 2)(x + 2)(x^2 + 4)$

5. Expand: $(a + b)(a^2 - ab + b^2)$

ANSWER $a^3 + b^3$

6. Make x the subject: $a(x - 1) = b(x + 1)$, where $a \neq b$

ANSWER $x = \frac{a + b}{a - b}$

7. Factorise: $x^2 - 5x + 6$

ANSWER $(x - 2)(x - 3)$

07 Homework

Homework — 5 problems

Algebra 8, pp. 3–6 (revision exercises). Written work, about 20 minutes.

1. Simplify: $(2x^3)^2 \cdot 3x \div 4x^4$ (answer: $3x^3$)
2. Expand and simplify: $(2x - 1)(x + 5) - 2x^2$ (answer: $9x - 5$)
3. Factorise: $3x^2 - 12$
4. Factorise: $x^2 + 8x + 16$

5. Solve: $\frac{2x+1}{4} - \frac{x}{3} = 1$

Algebraic expressions

Whole expressions, fractional expressions, and the values of the variable that are simply not allowed.

LESSON 4–5

2 × 40 MIN

ALGEBRA 8, §1

STAGE 9 · 2.1–2.2

LESSON CLOCK



Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

LEARNING OBJECTIVES

- Tell a whole (integer) expression from a fractional one.
- Find the permissible values of the variable — the domain of an expression.
- Evaluate an algebraic expression for given values of its letters.
- Write a described quantity as an algebraic expression.

TEXTBOOK

National §1, pp. 7–11

Cambridge Learner's Book pp. 22–29

Workbook Workbook 2.1–2.2

01 Explanation

Two kinds of expression

An expression built from numbers and letters using addition, subtraction, multiplication and raising to a whole power — but **never dividing by a letter** — is called a **whole** or **integer expression**.

$$3x^2 - 5x + 1 \quad \frac{2a}{7} \quad (x - y)^3$$

If the expression divides by something containing a letter, it is a **fractional expression**.

$$\frac{5}{x} \quad \frac{x+1}{x-3} \quad 2 + \frac{1}{a}$$

Note the difference between $\frac{2a}{7}$ and $\frac{7}{2a}$. The first divides by the *number* 7 and is

whole; the second divides by $2a$ and is fractional. It is the letter in the denominator

that changes everything.

Permissible values of the variable

A whole expression has a value for *every* value of its letters. A fractional expression does not: division by zero has no meaning.

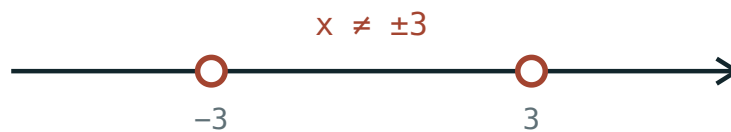
DEFINITION

The **permissible values** of the variable are all the values for which the expression has a value — that is, all values that do not make any denominator zero.

So for $\frac{x+1}{x-3}$ we must exclude $x = 3$, and we write $x \neq 3$.

To find them: set every denominator equal to zero, solve, and exclude those answers.

$$x^2 - 9 = 0 \implies x = 3 \text{ or } x = -3, \text{ so } x \neq \pm 3$$



The two excluded points are drawn as open circles — the line has holes there.

CAREFUL

Some denominators never vanish. $x^2 + 4$ is at least 4 for every real x , so $\frac{3}{x^2 + 4}$ is defined for **all** values of x .

The value of an expression

To evaluate, substitute and calculate — but check the value is permissible first.

$\frac{a^2 - b^2}{a - b}$ at $a = 7, b = 3$: the denominator is $7 - 3 = 4 \neq 0$, so the value is $\frac{49 - 9}{4} = 10$.

At $a = 3, b = 3$ the denominator is 0 — the expression has no value there at all.

Writing expressions from words

- “a number 5 more than three times x ” $\rightarrow 3x + 5$
- “the reciprocal of $x - 5$ ” $\rightarrow \frac{1}{x - 5}$, with $x \neq 5$
- “the average speed for s km covered in t hours” $\rightarrow \frac{s}{t}$, with $t \neq 0$

02 Worked examples

EXAMPLE 1 Find the permissible values of $\frac{x + 1}{x^2 - 5x + 6}$

① $x^2 - 5x + 6 = 0$
Set the denominator to zero.

② $(x - 2)(x - 3) = 0$
Factorise.

③ $x = 2$ or $x = 3$
These are the forbidden values.

④ $x \neq 2, x \neq 3$
Everything else is permitted.

ANSWER

$x \neq 2$ and $x \neq 3$

EXAMPLE 2 For which x is $\frac{x-2}{x+3}$ equal to zero?

① A fraction is zero exactly when its **numerator** is zero and its denominator is not.
Both conditions matter.

② $x - 2 = 0 \implies x = 2$
Numerator condition.

③ at $x = 2$ the denominator is $5 \neq 0$ ✓
Check the value is permissible.

ANSWER

$$x = 2$$

03 Interactive model

Slide x across the excluded value and let the class watch the fraction break.

Try a value of x

$$\frac{x+1}{x^2-4}$$

x 3

denominator 5

value 0.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic expression	Algebraik ifoda	Алгебраическое выражение
Whole (integer) expression	Butun ifoda	Целое выражение
Fractional expression	Kasr ifoda	Дробное выражение
Variable	O'zgaruvchi	Переменная
Permissible values	Mumkin bo'lgan qiymatlar	Допустимые значения
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Value of an expression	Ifodaning qiymati	Значение выражения
Undefined	Aniqlanmagan	Не определено
Reciprocal	Teskari son	Обратное число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

Which of these is a **fractional** expression?

$$\frac{3x}{5}$$

$$\frac{5}{3x}$$

$$3x - 5$$

$$(3x)^2$$

The permissible values of $\frac{7}{x+2}$ are:

$$x \neq 2$$

$$x \neq -2$$

$$x \neq 0$$

all values of x

For which x is $\frac{4}{x^2+1}$ undefined?

$$x = 1$$

$$x = -1$$

$$x = \pm 1$$

never — it is defined for all x

The value of $\frac{x^2 - 4}{x + 2}$ at $x = 3$ is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Which of $3x/5$, $5/x$, $x/2 + 1$, $7/(x - 1)$ are fractional expressions?

ANSWER $5/x$ and $7/(x - 1)$

2. Permissible values of $\frac{1}{x}$

ANSWER $x \neq 0$

3. Permissible values of $\frac{1}{x - 4}$

ANSWER $x \neq 4$

4. Permissible values of $\frac{1}{x + 7}$

ANSWER $x \neq -7$

5. Value of $\frac{x + 3}{x - 1}$ when $x = 5$

ANSWER 2

6. Value of $\frac{2a}{a + 2}$ when $a = 4$

ANSWER $\frac{4}{3}$

7. Permissible values of $\frac{3}{2x}$

ANSWER $x \neq 0$

Medium · the standard question

7 PROBLEMS

1. Permissible values of $\frac{x+1}{x^2-9}$

ANSWER $x \neq 3, x \neq -3$

2. Permissible values of $\frac{2x}{x^2+1}$

ANSWER All values of x — the denominator is never 0.

3. Permissible values of $\frac{5}{x^2-4x}$

ANSWER $x \neq 0, x \neq 4$

4. Value of $\frac{a^2-b^2}{a-b}$ when $a = 7, b = 3$

ANSWER 10

5. Value of $\frac{x^2-4}{x+2}$ when $x = 3$

ANSWER 1

6. Write the reciprocal of $x-5$ and state its permissible values

ANSWER $\frac{1}{x-5}, x \neq 5$

7. For which x is $\frac{x-2}{x+3} = 0$?

ANSWER $x = 2$

Hard · stretch and reason

7 PROBLEMS

1. Permissible values of $\frac{x-1}{x^2-5x+6}$

ANSWER $x \neq 2, x \neq 3$

2. Permissible values of $\frac{1}{x^2+x}$

ANSWER $x \neq 0, x \neq -1$

3. Solve $\frac{x^2-9}{x-3} = 0$

ANSWER $x = -3$ only — $x = 3$ is not permissible.

4. Simplify, then evaluate $\frac{x^2-6x+9}{x-3}$ at $x = 10$

ANSWER $x - 3 = 7$

5. Permissible values of $\frac{3}{|x|-2}$

ANSWER $x \neq 2, x \neq -2$

6. Permissible values of $\frac{5}{x^2-7x+10}$

ANSWER $x \neq 2, x \neq 5$

7. For which x is $\frac{x+4}{x^2+4}$ undefined?

ANSWER For no value — $x^2 + 4 \geq 4$ always.

07 Homework

Homework — 5 problems

Algebra 8, §1, pp. 7–11. Cambridge Workbook 2.1 for the substitution practice.

1. State the permissible values of $\frac{x+5}{x^2-16}$
2. State the permissible values of $\frac{2}{x^2+3x}$
3. Find the value of $\frac{a+b}{a-b}$ when $a=9, b=4$
4. For which x is $\frac{x+6}{x-1}$ equal to zero?
5. A car covers s km in t hours. Write its average speed as an algebraic expression and state the permissible values of t .

Algebraic fraction.

Cancelling fractions

The fundamental property of a fraction, and the single rule that governs every simplification in this chapter: factorise first, cancel second.

LESSON 6–8

3 × 40 MIN

ALGEBRA 8, §2

STAGE 9 · 2.5

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State what an algebraic fraction is and give its permissible values.
- Use the fundamental property of a fraction in both directions.
- Cancel a fraction by factorising the numerator and the denominator.
- Handle the sign rules, including the swap $b - a = -(a - b)$.

TEXTBOOK

National	§2, pp. 12–17
Cambridge	Learner’s Book pp. 40–44
Workbook	Workbook 2.5

01

Explanation

What an algebraic fraction is

DEFINITION

An expression of the form $\frac{A}{B}$, where A and B are polynomials and B is not the zero polynomial, is called an **algebraic fraction**. A is the numerator, B the denominator.

It is defined only for values that keep $B \neq 0$. Every answer in this chapter carries that condition, whether or not the question asks for it.

The fundamental property

THE RULE EVERYTHING RESTS ON

$$\frac{A}{B} = \frac{A \cdot C}{B \cdot C}, \text{ where } C \neq 0$$

Multiplying — or dividing — the numerator and the denominator by the same non-zero expression does not change the fraction.

Read left to right it lets you **build up** a fraction to a required denominator, which is what you need for addition. Read right to left it lets you **cancel**, which is what you need now. It is the same rule both ways.

Cancelling — the method

1. **Factorise** the numerator completely.
2. **Factorise** the denominator completely.
3. Cancel every factor common to both.
4. State the permissible values.

$$\frac{x^2 - 9}{x + 3} = \frac{(x - 3)(x + 3)}{x + 3} = x - 3, \quad x \neq -3$$

THE ONE MISTAKE THAT COSTS THE MOST MARKS

You may cancel **factors**, never **terms**. $\frac{x + 3}{3}$ is **not** x — the 3 in the numerator is added, not

multiplied. If you cannot see a bracket around the whole numerator, you cannot cancel.

Signs, and the swap that saves you

A minus sign may sit in three places, and all three mean the same thing:

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}$$

The move you will use constantly:

$$b - a = -(a - b)$$

So $\frac{3a - 3b}{b - a} = \frac{3(a - b)}{-(a - b)} = -3$. Whenever a bracket appears reversed, pull out a minus sign and the factors match.

02 Worked examples

EXAMPLE 1 *Cancel* $\frac{x^2 - 6x + 9}{x^2 - 9}$

① $x^2 - 6x + 9 = (x - 3)^2$

Numerator: perfect square.

② $x^2 - 9 = (x - 3)(x + 3)$

Denominator: difference of two squares.

③ $\frac{(x - 3)(x - 3)}{(x - 3)(x + 3)}$

Now the common factor is visible.

④ $\frac{x - 3}{x + 3}$

Cancel one $(x - 3)$ from top and bottom.

⑤ $x \neq 3, x \neq -3$

Both zeros of the original denominator stay excluded.

ANSWER

$$\frac{x - 3}{x + 3}, x \neq \pm 3$$

EXAMPLE 2 *Cancel* $\frac{2x^2 + 7x + 3}{x^2 - 9}$

① $2 \cdot 3 = 6$, and $1 + 6 = 7$

Split the middle term of the numerator.

② $2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$

Group and factor.

③ $x^2 - 9 = (x - 3)(x + 3)$

Denominator.

④ $\frac{(2x + 1)(x + 3)}{(x - 3)(x + 3)} = \frac{2x + 1}{x - 3}$

Cancel $(x + 3)$.

ANSWER

$$\frac{2x + 1}{x - 3}, \quad x \neq \pm 3$$

EXAMPLE 3 *Cancel* $\frac{m^2 - mn}{n^2 - mn}$

① $m^2 - mn = m(m - n)$

Common factor m.

② $n^2 - mn = n(n - m)$

Common factor n.

③ $n - m = -(m - n)$

Reverse the bracket so it matches the numerator.

④ $\frac{m(m - n)}{-n(m - n)} = -\frac{m}{n}$

Cancel (m - n); the minus sign stays.

ANSWER

$-\frac{m}{n}, n \neq 0, m \neq n$

03 Interactive model

Do the first one with the class; let a learner predict each next step before you reveal it.

Cancelling step by step

$$\frac{x^2 - 9}{x + 3}$$

① $x^2 - 9 = (x - 3)(x + 3)$

Difference
of
two
squares.

②
$$\frac{(x - 3)(x + 3)}{x + 3}$$

The
denominator
is
already
a
single
factor.

③ $x - 3$

Cancel
(x +
3)
—
but
remember
 $x \neq$
-3.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Fundamental property of a fraction	Kasrning asosiy xossasi	Основное свойство дроби
Cancel (reduce) a fraction	Kasrni qisqartirish	Сокращение дроби
Factor	Ko'paytuvchi	Множитель
Term	Had	Слагаемое
Polynomial	Ko'phad	Многочлен
Identical transformation	Ayniy almashtirish	Тождественное преобразование
Sign	Ishora	Знак
Bracket	Qavs	Скобка
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x + 3$ $\div 3$ simplifies to:

$$x$$

$$x + 1$$

it does not simplify

$$\frac{x}{3}$$

$6a$ $\div 9a$ equals:

$$\frac{2}{3}$$

$$\frac{2a}{3}$$

$$\frac{6}{9a}$$

$$3$$

$x^2 - 4x$ $x - 4$ equals:

$$x - 4$$

$$x$$

$$4x$$

$$\frac{x}{4}$$

$b - a$ is the same as:

$$a - b$$

$$-(a - b)$$

$$-(b - a)$$

$$(a - b)^2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Cancel:* $\frac{6a}{9a}$

ANSWER $\frac{2}{3}, a \neq 0$

2. *Cancel:* $\frac{12x^2}{4x}$

ANSWER $3x, x \neq 0$

3. *Cancel:* $\frac{5ab}{10a}$

ANSWER $\frac{b}{2}, a \neq 0$

4. *Cancel:* $\frac{x^2}{x^5}$

ANSWER $\frac{1}{x^3}, x \neq 0$

5. *Cancel:* $\frac{3(x+1)}{6(x+1)}$

ANSWER $\frac{1}{2}, x \neq -1$

6. *Cancel:* $\frac{a+b}{2(a+b)}$

ANSWER $\frac{1}{2}, a \neq -b$

7. *Cancel:* $\frac{-4x}{8x}$

ANSWER $-\frac{1}{2}, x \neq 0$

Medium · the standard question

7 PROBLEMS

1. *Cancel:* $\frac{x^2 - 9}{x + 3}$

ANSWER $x - 3, x \neq -3$

2. *Cancel:* $\frac{x^2 - 4x}{x - 4}$

ANSWER $x, x \neq 4$

3. *Cancel:* $\frac{2x + 6}{x^2 - 9}$

ANSWER $\frac{2}{x - 3}, x \neq \pm 3$

4. *Cancel:* $\frac{a^2 - b^2}{a^2 + 2ab + b^2}$

ANSWER $\frac{a - b}{a + b}, a \neq -b$

5. *Cancel:* $\frac{x^2 - 6x + 9}{x^2 - 9}$

ANSWER $\frac{x - 3}{x + 3}, x \neq \pm 3$

6. *Cancel:* $\frac{5x - 5y}{x^2 - y^2}$

ANSWER $\frac{5}{x + y}, x \neq \pm y$

7. *Cancel:* $\frac{3a - 3b}{b - a}$

ANSWER $-3, a \neq b$

Hard · stretch and reason

7 PROBLEMS

1. *Cancel:* $\frac{x^3 - x}{x^2 + x}$

ANSWER $x - 1, x \neq 0, -1$

2. *Cancel:* $\frac{x^2 - 5x + 6}{x^2 - 4}$

ANSWER $\frac{x - 3}{x + 2}, x \neq \pm 2$

3. *Cancel:* $\frac{2x^2 + 7x + 3}{x^2 - 9}$

ANSWER $\frac{2x + 1}{x - 3}, x \neq \pm 3$

4. *Cancel:* $\frac{x^2 - 2x}{x^2 - 4}$

ANSWER $\frac{x}{x + 2}, x \neq \pm 2$

5. *Cancel:* $\frac{x^4 - 16}{x^2 + 4}$

ANSWER $x^2 - 4$

6. *Cancel:* $\frac{m^2 - mn}{n^2 - mn}$

ANSWER $-\frac{m}{n}, n \neq 0, m \neq n$

7. *Cancel:* $\frac{x^2 + x - 6}{x^2 - 4x + 4}$

ANSWER $\frac{x + 3}{x - 2}, x \neq 2$

07 Homework**Homework — 6 problems**

Algebra 8, §2, pp. 12–17. State the permissible values with every answer.

1. *Cancel:* $\frac{8xy}{12x}$

2. *Cancel:* $\frac{x^2 - 25}{x - 5}$

3. *Cancel:* $\frac{3x + 12}{x^2 - 16}$

4. *Cancel:* $\frac{x^2 + 4x + 4}{x^2 - 4}$

5. *Cancel:* $\frac{7a - 7b}{b - a}$

6. Explain in one sentence why $\frac{x + 5}{5}$ cannot be cancelled.

Reducing fractions to a common denominator

Building fractions up instead of cutting them down — the fundamental property read from left to right.

LESSON 9–11

3 × 40 MIN

ALGEBRA 8, §3

STAGE 9 · 2.5

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Factorise denominators and build the lowest common denominator from the factors.
- Find the additional factor for each fraction.
- Rewrite two or three fractions with the same denominator.
- Recognise the reversed-bracket case $a - b$ and $b - a$.

TEXTBOOK

National	§3, pp. 18–21
Cambridge	Learner’s Book pp. 40–44
Workbook	Workbook 2.5

01

Explanation

Why a common denominator at all

You cannot add $\frac{1}{2}$ and $\frac{1}{3}$ directly because the pieces are different sizes. Cut both into sixths and the addition becomes counting. Algebraic fractions behave exactly the same way — only the “sixths” are now expressions.

The lowest common denominator

METHOD

- Factorise every denominator completely.
- Take **every factor that appears anywhere**, each to its **highest** power.
- The product of these is the lowest common denominator (LCD).
- Divide the LCD by each denominator — that quotient is the fraction's **additional factor**.
- Multiply the numerator and denominator of each fraction by its additional factor.

Numbers work the same way. For $\frac{1}{4x}$ and $\frac{1}{6y}$: the numerical LCM of 4 and 6 is

12, the letters give xy , so the LCD is $12xy$.

A worked pattern

Bring $\frac{3}{x^2 - 9}$ and $\frac{1}{x - 3}$ to a common denominator.

$$x^2 - 9 = (x - 3)(x + 3), \quad x - 3 = (x - 3)$$

Every factor that appears: $(x - 3)$ and $(x + 3)$, each to the first power. So

$$LCD = (x - 3)(x + 3)$$

The first fraction already has it — additional factor 1. The second needs $(x + 3)$:

$$\frac{3}{(x - 3)(x + 3)} \quad \text{and} \quad \frac{x + 3}{(x - 3)(x + 3)}$$

DO NOT MULTIPLY BLINDLY

Multiplying the denominators together always *works*, but it can give $(x - 3)(x + 3)(x - 3)$ where $(x - 3)(x + 3)$ was enough — and then you are cancelling it back down at the end. Factorise first, and the lowest denominator appears by itself.

When one denominator is the reverse of the other

$x - y$ and $y - x$ are not different denominators — they differ by a sign. Multiply the second fraction's numerator and denominator by -1 :

$$\frac{y}{y - x} = \frac{-y}{x - y}$$

Now both fractions share the denominator $x - y$ and no extra factors are needed.

02 Worked examples

EXAMPLE 1 Bring $\frac{1}{x^2 - 4}$ and $\frac{1}{x^2 - 4x + 4}$ to a common denominator

① $x^2 - 4 = (x - 2)(x + 2)$

First denominator.

② $x^2 - 4x + 4 = (x - 2)^2$

Second denominator — note the square.

③ $LCD = (x - 2)^2(x + 2)$

$(x - 2)$ appears to the power 2, so take it squared.

④ additional factors: $(x - 2)$ and $(x + 2)$

$LCD \div$ each denominator.

⑤ $\frac{x - 2}{(x - 2)^2(x + 2)}$ and $\frac{x + 2}{(x - 2)^2(x + 2)}$

Multiply top and bottom by them.

ANSWER

$$LCD = (x - 2)^2(x + 2), \quad x \neq \pm 2$$

EXAMPLE 2 Find the LCD of $\frac{2}{3a^2b}$ and $\frac{5}{6ab^3}$

① $LCM(3, 6) = 6$

The numbers first.

② a : highest power a^2

a^2 beats a .

③ b : highest power b^3

b^3 beats b .

④ $LCD = 6a^2b^3$

Multiply what you collected.

ANSWER

$$6a^2b^3$$

03 Interactive model

Ask the class for the LCD before you open the table — then check their answer against it.

Building the common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
$x^2 - 1$	$(x - 1)(x + 1)$	1
$x + 1$	$(x + 1)$	$(x - 1)$

LOWEST COMMON DENOMINATOR

$$(x - 1)(x + 1) = x^2 - 1$$

The first denominator already *is* the LCD — only the second has to be built up.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Additional factor	Qo'shimcha ko'paytuvchi	Дополнительный множитель
Lowest common multiple	EKUK	НОК
Highest power	Eng yuqori daraja	Наивысшая степень
Bring to a common denominator	Umumiy maxrajga keltirish	Привести к общему знаменателю
Factorised form	Ko'paytma ko'rinishi	Разложенный вид
Reversed bracket	Teskari qavs	Перевернутая скобка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The LCD of $\frac{1}{x}$ and $\frac{1}{x^2}$ is:

x

x^2

x^3

$2x$

The LCD of $\frac{1}{x-1}$ and $\frac{1}{x+1}$ is:

$x^2 - 1$

$x^2 + 1$

x

$(x - 1)^2$

To write $\frac{3}{x}$ with denominator x^2 , the numerator becomes:

3

$3x$

$3x^2$

x

y $y - x$ written with denominator $x - y$ is:

$$\frac{y}{x - y}$$

$$\frac{-y}{x - y}$$

$$\frac{x}{x - y}$$

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. LCD of $\frac{1}{2}$ and $\frac{1}{3}$

ANSWER 6

2. LCD of $\frac{1}{x}$ and $\frac{1}{y}$

ANSWER xy

3. LCD of $\frac{1}{2a}$ and $\frac{1}{3a}$

ANSWER $6a$

4. LCD of $\frac{1}{x}$ and $\frac{1}{x^2}$

ANSWER x^2

5. LCD of $\frac{1}{x-1}$ and $\frac{1}{x+1}$

ANSWER $(x-1)(x+1)$

6. Write $\frac{3}{x}$ with denominator x^2

ANSWER $\frac{3x}{x^2}$

7. Write $\frac{2}{a}$ with denominator ab

ANSWER $\frac{2b}{ab}$

Medium · the standard question

7 PROBLEMS

1. LCD of $\frac{1}{x^2 - 1}$ and $\frac{1}{x + 1}$

ANSWER $(x - 1)(x + 1)$

2. LCD of $\frac{1}{2x}$ and $\frac{1}{3x^2}$

ANSWER $6x^2$

3. Bring $\frac{1}{x - 2}$ and $\frac{1}{x + 2}$ to a common denominator

ANSWER $\frac{x + 2}{x^2 - 4}$ and $\frac{x - 2}{x^2 - 4}$

4. LCD of $\frac{1}{x^2 + 2x}$ and $\frac{1}{x + 2}$

ANSWER $x(x + 2)$

5. LCD of $\frac{1}{a^2 - b^2}$ and $\frac{1}{a - b}$

ANSWER $(a - b)(a + b)$

6. Bring $\frac{3}{x^2 - 9}$ and $\frac{1}{x - 3}$ to a common denominator

ANSWER $\frac{3}{x^2 - 9}$ and $\frac{x + 3}{x^2 - 9}$

7. LCD of $\frac{1}{4x}$ and $\frac{1}{6y}$

ANSWER $12xy$

Hard · stretch and reason

7 PROBLEMS

1. LCD of $\frac{1}{x^2 - 4}$ and $\frac{1}{x^2 - 4x + 4}$

ANSWER $(x - 2)^2(x + 2)$

2. LCD of $\frac{1}{x^2 + x}$ and $\frac{1}{x^2 - x}$

ANSWER $x(x - 1)(x + 1)$

3. LCD of $\frac{1}{x^2 + 5x + 6}$ and $\frac{1}{x^2 + 4x + 4}$

ANSWER $(x + 2)^2(x + 3)$

4. Bring $\frac{1}{x^2 - 1}$ and $\frac{1}{x^2 + 2x + 1}$ to a common denominator

ANSWER LCD $(x - 1)(x + 1)^2$: $\frac{x + 1}{(x - 1)(x + 1)^2}$ and $\frac{x - 1}{(x - 1)(x + 1)^2}$

5. LCD of $\frac{2}{3a^2b}$ and $\frac{5}{6ab^3}$

ANSWER $6a^2b^3$

6. Bring $\frac{x}{x - y}$ and $\frac{y}{y - x}$ to a common denominator

ANSWER $\frac{x}{x - y}$ and $\frac{-y}{x - y}$

7. LCD of $\frac{1}{x^2 - 4x}$ and $\frac{1}{x - 4}$

ANSWER $x(x - 4)$

07 Homework

Homework — 5 problems

Algebra 8, §3, pp. 18–21. Factorise every denominator before you write the LCD.

1. Find the LCD of $\frac{1}{x^2 - 25}$ and $\frac{1}{x + 5}$
2. Find the LCD of $\frac{1}{6ab}$ and $\frac{1}{9b^2}$
3. Bring $\frac{2}{x - 3}$ and $\frac{5}{x + 3}$ to a common denominator
4. Bring $\frac{1}{x^2 + x}$ and $\frac{1}{x}$ to a common denominator
5. Write $\frac{a}{a - b}$ and $\frac{b}{b - a}$ with the same denominator

Addition and subtraction of algebraic fractions

Same denominator — add the numerators. Different denominators — build them the same first, then add.

LESSON 12–14

3 × 40 MIN

ALGEBRA 8, §4

STAGE 9 · 2.5

LESSON CLOCK

5'	12'	8'	13'	2'
----	-----	----	-----	----

Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

LEARNING OBJECTIVES

- Add and subtract fractions with the same denominator.
- Add and subtract fractions with different denominators using the LCD.
- Handle subtraction correctly — bracket the whole numerator being subtracted.
- Simplify the result and state the permissible values.

TEXTBOOK

National \$4, pp. 22–26

Cambridge Learner's Book pp. 40–44

Workbook Workbook 2.5

01 Explanation

Same denominator

$$\frac{A}{C} \pm \frac{B}{C} = \frac{A \pm B}{C}$$

Keep the denominator, combine the numerators, then **always try to cancel the result** — the answer is often much simpler than the working suggests.

$$\frac{x}{x+3} + \frac{3}{x+3} = \frac{x+3}{x+3} = 1$$

Different denominators

FOUR STEPS, EVERY TIME

1. Factorise the denominators and find the LCD.
2. Find each additional factor and build both fractions up.

3. Combine the numerators over the single denominator.
4. Simplify the numerator, cancel if you can, state the permissible values.

$$\frac{1}{x-1} + \frac{1}{x+1} = \frac{(x+1) + (x-1)}{(x-1)(x+1)} = \frac{2x}{x^2-1}$$

The bracket that everyone forgets

SUBTRACTION

When you subtract a fraction, the minus sign belongs to the **whole** numerator, so put it in a bracket before you open it.

$$\frac{2x}{x^2-4} - \frac{1}{x-2} = \frac{2x - (x+2)}{(x-2)(x+2)} = \frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2}$$

Writing $2x - x + 2$ instead of $2x - (x + 2)$ changes the answer completely.

A whole expression is a fraction too

Any expression can be written over the denominator 1, so it can join the sum:

$$1 - \frac{1}{x+1} = \frac{(x+1) - 1}{x+1} = \frac{x}{x+1}$$

02 Worked examples

EXAMPLE 1 *Simplify* $\frac{5}{x^2 - 9} + \frac{1}{x + 3}$

① $x^2 - 9 = (x - 3)(x + 3)$

Factorise.

② $LCD = (x - 3)(x + 3)$

The second denominator is one of its factors.

③
$$\frac{5}{(x - 3)(x + 3)} + \frac{x - 3}{(x - 3)(x + 3)}$$

Additional factor $(x - 3)$ for the second fraction.

④
$$\frac{5 + x - 3}{(x - 3)(x + 3)} = \frac{x + 2}{x^2 - 9}$$

Combine and tidy the numerator.

ANSWER

$$\frac{x + 2}{x^2 - 9}, \quad x \neq \pm 3$$

EXAMPLE 2 *Simplify* $\frac{1}{x-1} - \frac{2}{x^2-1}$

① $x^2 - 1 = (x - 1)(x + 1)$

Factorise.

② $LCD = (x - 1)(x + 1)$

The first denominator needs the factor $(x + 1)$.

③ $\frac{x+1}{(x-1)(x+1)} - \frac{2}{(x-1)(x+1)}$

Build the first fraction up.

④ $\frac{(x+1)-2}{(x-1)(x+1)} = \frac{x-1}{(x-1)(x+1)}$

Combine — keep the bracket.

⑤ $\frac{1}{x+1}$

Cancel $(x - 1)$. Always look for this last step.

ANSWER

$$\frac{1}{x+1}, \quad x \neq \pm 1$$

EXAMPLE 3 *Simplify* $\frac{1}{x(x+1)} + \frac{1}{(x+1)(x+2)}$

① $LCD = x(x+1)(x+2)$
 $(x+1)$ is shared, so it is taken once.

② $\frac{x+2}{x(x+1)(x+2)} + \frac{x}{x(x+1)(x+2)}$
 Additional factors $(x+2)$ and x .

③ $\frac{2x+2}{x(x+1)(x+2)}$
 Combine the numerators.

④ $\frac{2(x+1)}{x(x+1)(x+2)} = \frac{2}{x(x+2)}$
 Factorise the numerator and cancel $(x+1)$.

ANSWER

$$\frac{2}{x(x+2)}, \quad x \neq 0, -1, -2$$

03 Interactive model

Reveal the four steps one at a time and ask which step the class thinks comes next.

Adding and subtracting, step by step

$$\frac{1}{x-1} + \frac{1}{x+1}$$

① $LCD = (x-1)(x+1)$

Different
factors,
so
multiply
them.

② *additional factors: $(x+1)$ and $(x-1)$*

LCD
÷
each
denominator.

③
$$\frac{(x+1) + (x-1)}{(x-1)(x+1)}$$

Build
both
up
and
combine.

④
$$\frac{2x}{x^2-1}$$

Tidy
the
numerator.
Nothing
cancels
here.

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Add	Qo‘shish	Сложение
Subtract	Ayirish	Вычитание
Like fractions	Bir xil maxrajli kasrlar	Дроби с одинаковыми знаменателями
Unlike fractions	Har xil maxrajli kasrlar	Дроби с разными знаменателями
Combine	Birlashtirish	Объединить
Numerator	Surat	Числитель
Whole expression	Butun ifoda	Целое выражение
Simplify the result	Natijani soddalashtirish	Упростить результат

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

3 x + 5 x equals:

$$\frac{8}{2x}$$

$$\frac{8}{x}$$

$$\frac{15}{x}$$

$$\frac{8}{x^2}$$

2x x² - 4 - 1 x - 2 equals:

$$\frac{1}{x+2}$$

$$\frac{2x-1}{x^2-4}$$

$$\frac{x+2}{x^2-4}$$

$$\frac{3x+2}{x^2-4}$$

1 $a - b$ + 1 $b - a$ equals:

$$\frac{2}{a - b}$$

$$0$$

$$\frac{2}{b - a}$$

it cannot be combined

x $x + 3$ + 3 $x + 3$ equals:

$$\frac{x + 3}{2x + 6}$$

$$1$$

$$\frac{3x}{x + 3}$$

$$x$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{3}{x} + \frac{5}{x}$

ANSWER $\frac{8}{x}$

2. $\frac{7}{a} - \frac{2}{a}$

ANSWER $\frac{5}{a}$

3. $\frac{x}{4} + \frac{x}{4}$

ANSWER $\frac{x}{2}$

4. $\frac{2a}{b} + \frac{3a}{b}$

ANSWER $\frac{5a}{b}$

5. $\frac{1}{2} + \frac{1}{x}$

ANSWER $\frac{x+2}{2x}$

6. $\frac{1}{x} - \frac{1}{y}$

ANSWER $\frac{y-x}{xy}$

7.
$$\frac{3}{x+1} + \frac{2}{x+1}$$

ANSWER
$$\frac{5}{x+1}$$

Medium · the standard question

7 PROBLEMS

$$1. \quad \frac{1}{x-1} + \frac{1}{x+1}$$

ANSWER $\frac{2x}{x^2-1}$

$$2. \quad \frac{1}{x-2} - \frac{1}{x+2}$$

ANSWER $\frac{4}{x^2-4}$

$$3. \quad \frac{x}{x+3} + \frac{3}{x+3}$$

ANSWER 1, $x \neq -3$

$$4. \quad \frac{2}{x} + \frac{3}{x^2}$$

ANSWER $\frac{2x+3}{x^2}$

$$5. \quad \frac{a}{a-b} - \frac{b}{a-b}$$

ANSWER 1, $a \neq b$

$$6. \quad \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

ANSWER $\frac{bc+ac+ab}{abc}$

$$7. \quad \frac{5}{x^2-9} + \frac{1}{x+3}$$

ANSWER $\frac{x+2}{x^2-9}$

Hard · stretch and reason

7 PROBLEMS

$$1. \quad \frac{1}{x^2 - x} + \frac{1}{x^2 + x}$$

ANSWER $\frac{2}{x^2 - 1}$

$$2. \quad \frac{1}{x - 3} + \frac{1}{x + 3}$$

ANSWER $\frac{2x}{x^2 - 9}$

$$3. \quad \frac{1}{a - b} + \frac{1}{b - a}$$

ANSWER $0, a \neq b$

$$4. \quad \frac{2x}{x^2 - 4} - \frac{1}{x - 2}$$

ANSWER $\frac{1}{x + 2}$

$$5. \quad 1 - \frac{1}{x + 1}$$

ANSWER $\frac{x}{x + 1}$

$$6. \quad \frac{1}{x - 1} - \frac{2}{x^2 - 1}$$

ANSWER $\frac{1}{x + 1}$

$$7. \quad \frac{1}{x(x + 1)} + \frac{1}{(x + 1)(x + 2)}$$

ANSWER $\frac{2}{x(x + 2)}$

07 Homework

Homework — 6 problems

Algebra 8, §4, pp. 22–26. Cancel the answer wherever you can.

$$1. \quad \frac{4}{x} + \frac{9}{x}$$

$$2. \quad \frac{1}{x+2} + \frac{1}{x-2}$$

$$3. \quad \frac{3}{x-4} - \frac{3}{x+4}$$

$$4. \quad \frac{x}{x-5} - \frac{5}{x-5}$$

$$5. \quad \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$$

$$6. \quad 2 - \frac{3}{x-1}$$

Control work 1 · Algebraic fractions

The first assessment of the quarter, followed by a full lesson on the mistakes it produced.

LESSON 15–16

2 × 40 MIN

ALGEBRA 8, §§1–4

STAGE 9 · UNIT 2 CHECK

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess: permissible values, cancelling, common denominators, addition and subtraction.
- Work through the errors the paper reveals, by type rather than by question.
- Re-solve every task that was lost, correctly and in full.

TEXTBOOK

National	Revision of §§1–4
Cambridge	Learner’s Book pp. 21–54
Workbook	Workbook Unit 2

01

Explanation

Lesson 15 — the paper (40 minutes)

Two variants of seven tasks, printed below as the Easy and Medium practice sets. Both variants carry the same marks and cover the same four skills:

- tasks 1–2 · permissible values (§1)
- tasks 3–4 · cancelling a fraction (§2)
- task 5 · the lowest common denominator (§3)
- tasks 6–7 · addition and subtraction (§4)

Marking: 2 marks per task, 14 in total. A mark is lost for a missing permissible-value condition even when the algebra is right — say this before the paper starts, not after.

Lesson 16 — work on mistakes (40 minutes)

Do not go through the paper question by question. Sort the errors into the four types below, deal with each type once, then have every learner re-solve only their own lost tasks.

THE FOUR ERRORS THAT ACCOUNT FOR ALMOST EVERYTHING

1. **Cancelling a term instead of a factor** — $\frac{x+5}{5}$ written as x .
2. **The missing bracket in a subtraction** — $2x - x + 2$ instead of $2x - (x + 2)$.
3. **The permissible values dropped**, or taken from the *simplified* denominator instead of the original one.
4. **Multiplying the denominators together** without factorising, then failing to cancel back.

The Hard set below is built entirely from these four errors: each task is a piece of wrong working to be found and repaired. That is a better use of the hour than a second full paper.

02 Worked examples

EXAMPLE 1 Find and repair the error: $\frac{x+4}{4} = x$

- ① The 4 on top is a **term** — it is added to x , not multiplied by it.
Only factors cancel.
- ② Write the numerator as a product to test it: $x + 4$ has no factor 4.
So there is nothing to cancel.
- ③ $\frac{x+4}{4} = \frac{x}{4} + 1$
You may split it, but you may not cancel it.

ANSWER

The expression does not simplify to x . A correct alternative form is $\frac{x}{4} + 1$.

EXAMPLE 2**Find and repair the error:**

$$\frac{3x}{x^2 - 1} - \frac{1}{x - 1} = \frac{3x - 1}{x^2 - 1}$$

- ① The second fraction was not built up to the LCD.
Its denominator is $(x - 1)$, not $(x - 1)(x + 1)$.

- ② *additional factor $(x + 1)$*
So the numerator becomes $(x + 1)$, not 1.

- ③
$$\frac{3x - (x + 1)}{(x - 1)(x + 1)} = \frac{2x - 1}{x^2 - 1}$$

Both errors — the factor and the bracket — corrected.

ANSWER

$$\frac{2x - 1}{x^2 - 1}, \quad x \neq \pm 1$$

03 Interactive model

Use this in lesson 16: put a piece of wrong working on the board and let the class find the error before you reveal it.

Find the mistake

Claimed: $\frac{x+5}{5} = x$

- ① The 5 in the numerator is added, not multiplied.

Cancelling
needs
a
common
factor.

- ② Test it with a number: at

$x = 5$, the left side is

$\frac{10}{5}$, the right side is 5.
 $= 2$

One
counter-
example
settles
it.

- ③ Correct: the fraction does not simplify. It may be split as

$\frac{x}{5}$
 $+ 1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Variant	Variant	Вариант
Mark	Ball	Балл
Check the answer	Javobni tekshirish	Проверить ответ
Error	Xato	Ошибка
Correct	To'g'ri	Верно
Counter-example	Qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

A learner writes $\frac{x+7}{7} = x$. The error is:

In $\frac{4x}{x^2-1} - \frac{1}{x-1}$ the numerator after combining is:

After simplifying $\frac{x^2-4}{x-2}$ to $x+2$, the condition is:

The lowest common denominator of $\frac{1}{x^2 - 4}$ and $\frac{1}{x + 2}$ is:

$$(x^2 - 4)(x + 2)$$

$$x^2 - 4$$

$$(x + 2)^2$$

$$x - 2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** State the permissible values of $\frac{x+2}{x-6}$

ANSWER $x \neq 6$

2. **Task 2.** State the permissible values of $\frac{4}{x^2-16}$

ANSWER $x \neq \pm 4$

3. **Task 3.** Cancel $\frac{15ab}{20a}$

ANSWER $\frac{3b}{4}, a \neq 0$

4. **Task 4.** Cancel $\frac{x^2-16}{x+4}$

ANSWER $x-4, x \neq -4$

5. **Task 5.** Find the LCD of $\frac{1}{x^2-4}$ and $\frac{1}{x-2}$

ANSWER $(x-2)(x+2)$

6. **Task 6.** $\frac{5}{x} + \frac{2}{x}$

ANSWER $\frac{7}{x}$

7. **Task 7.** $\frac{1}{x-3} + \frac{1}{x+3}$

ANSWER $\frac{2x}{x^2-9}$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** State the permissible values of $\frac{x-1}{x+5}$

ANSWER $x \neq -5$

2. **Task 2.** State the permissible values of $\frac{3}{x^2 - 5x}$

ANSWER $x \neq 0, x \neq 5$

3. **Task 3.** Cancel $\frac{18xy}{24y}$

ANSWER $\frac{3x}{4}, y \neq 0$

4. **Task 4.** Cancel $\frac{x^2 - 8x + 16}{x^2 - 16}$

ANSWER $\frac{x-4}{x+4}, x \neq \pm 4$

5. **Task 5.** Find the LCD of $\frac{1}{x^2 + 3x}$ and $\frac{1}{x + 3}$

ANSWER $x(x + 3)$

6. **Task 6.** $\frac{9}{2a} - \frac{3}{2a}$

ANSWER $\frac{3}{a}$

7. **Task 7.** $\frac{3x}{x^2 - 1} - \frac{1}{x - 1}$

ANSWER $\frac{2x - 1}{x^2 - 1}$

1. Find the error: $\frac{x+6}{6} = x$

ANSWER A term was cancelled. The expression equals $\frac{x}{6} + 1$; test with $x = 6$.

2. Find the error: $\frac{2x}{x^2-4} - \frac{1}{x-2} = \frac{2x-1}{x^2-4}$

ANSWER The second fraction was not built up. Correct answer $\frac{1}{x+2}$.

3. Find the error: $\frac{x^2-25}{x-5} = x+5$ for every x

ANSWER True only for $x \neq 5$; the condition comes from the original denominator.

4. Find the error: LCD of $\frac{1}{x^2-9}$ and $\frac{1}{x-3}$ is $(x^2-9)(x-3)$

ANSWER Not lowest. $LCD = x^2 - 9$.

5. Find the error: $\frac{1}{x} + \frac{1}{y} = \frac{1}{x+y}$

ANSWER Denominators are never added. Correct: $\frac{x+y}{xy}$; check with $x = y = 2$.

6. Find the error: $\frac{a-b}{b-a} = 1$

ANSWER The brackets are reversed: $b-a = -(a-b)$, so the value is -1 .

7. Find the error: $\frac{3}{x-2} - \frac{3}{x+2} = 0$

ANSWER Different denominators cannot cancel. Correct: $\frac{12}{x^2-4}$.

07 Homework

Homework after the work-on-mistakes lesson

Re-solve, in full, every task you lost marks on. Then these three.

1. Cancel $\frac{x^2 - 7x + 12}{x^2 - 9}$ and state the permissible values

2. $\frac{2}{x - 4} - \frac{2}{x + 4}$

3. Write out, in your own words, the four errors listed in this lesson and one example of each.

Multiplication and division of algebraic fractions

No common denominator needed — but factorise before you multiply, or you will multiply out work you are about to cancel.

LESSON 17–19

3 × 40 MIN

ALGEBRA 8, §5

STAGE 9 · 8.3–8.4

LESSON CLOCK



Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Multiply algebraic fractions and simplify the result.
- Divide by multiplying by the reciprocal.
- Cancel across the multiplication sign before multiplying out.
- Raise an algebraic fraction to a power.

TEXTBOOK

National	\$5, pp. 27–29
Cambridge	Learner's Book pp. 175–184
Workbook	Workbook 8.3–8.4

01 Explanation

The two rules

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{A \cdot C}{B \cdot D} \quad \frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C} = \frac{A \cdot D}{B \cdot C}$$

Multiply straight across. To divide, **turn the second fraction upside down and multiply**. Nothing else changes.

For division you need $C \neq 0$ as well as $B \neq 0$ and $D \neq 0$ — you cannot divide by a fraction whose numerator is zero.

Cancel first, multiply last

THE HABIT THAT MAKES THIS TOPIC EASY

Factorise every numerator and denominator, cancel **anything on top against anything on the bottom**, and only then write down what is left. If you multiply out first you create a degree-4 polynomial that you must then factorise again.

$$\frac{x^2 - 9}{x} \cdot \frac{x^2}{x + 3} = \frac{(x - 3)(x + 3)}{x} \cdot \frac{x \cdot x}{x + 3} = x(x - 3)$$

The $(x + 3)$ cancels diagonally, and one x cancels with the other denominator. Two cancellations, no expansion.

Powers of a fraction

$$\left(\frac{A}{B}\right)^n = \frac{A^n}{B^n}$$

$$\left(\frac{2x}{3y}\right)^3 = \frac{8x^3}{27y^3} \text{ — the index reaches the numbers as well as the letters.}$$

Dividing by a whole expression

A whole expression is a fraction with denominator 1, so dividing by $x + 1$ means multiplying by $\frac{1}{x + 1}$:

$$\frac{x^2 - 1}{x} : (x + 1) = \frac{(x - 1)(x + 1)}{x} \cdot \frac{1}{x + 1} = \frac{x - 1}{x}$$

02 Worked examples

EXAMPLE 1 *Simplify* $\frac{2a}{a+b} \div \frac{4a^2}{a^2-b^2}$

① $\frac{2a}{a+b} \cdot \frac{a^2-b^2}{4a^2}$

Turn the second fraction over.

② $a^2 - b^2 = (a - b)(a + b)$

Factorise before multiplying.

③ $\frac{2a \cdot (a - b)(a + b)}{(a + b) \cdot 4a^2}$

Now every factor is visible.

④ $\frac{a - b}{2a}$

Cancel $(a + b)$, and cancel $2a$ against $4a^2$.

ANSWER

$$\frac{a - b}{2a}, \quad a \neq 0, a \neq \pm b$$

EXAMPLE 2 *Simplify* $\frac{x^2 - 5x + 6}{x^2 - 4} \cdot \frac{x^2 + 2x}{x - 3}$

① $x^2 - 5x + 6 = (x - 2)(x - 3)$

First numerator.

② $x^2 - 4 = (x - 2)(x + 2)$

First denominator.

③ $x^2 + 2x = x(x + 2)$

Second numerator — common factor.

④ $\frac{(x - 2)(x - 3)}{(x - 2)(x + 2)} \cdot \frac{x(x + 2)}{x - 3}$

Everything factorised.

⑤ *cancel* $(x - 2)$, $(x + 2)$ and $(x - 3)$

Three cancellations, all diagonal.

ANSWER

$x, x \neq \pm 2, x \neq 3$

EXAMPLE 3 Simplify $(1 - \frac{1}{x}) : \frac{x-1}{x^2}$

① $1 - \frac{1}{x} = \frac{x-1}{x}$

Combine the bracket first.

② $\frac{x-1}{x} \cdot \frac{x^2}{x-1}$

Turn the divisor over.

③ *cancel $(x-1)$, and x against x^2*

Two cancellations.

④ x

A surprisingly simple answer — typical of this topic.

ANSWER

$x, x \neq 0, x \neq 1$

03 Interactive model

Show this on the board and let a learner drive it.

Multiplying and dividing, step by step

$$\frac{x^2 - 9}{x} \cdot \frac{x^2}{x + 3}$$

① $x^2 - 9 = (x - 3)(x + 3)$

Factorise
before
anything
else.

② $\frac{(x - 3)(x + 3)}{x} \cdot \frac{x \cdot x}{x + 3}$

Split
 x^2
so
the
cancelling
is
visible.

③ *cancel $(x + 3)$ and one x*

Both
cancellations
go
diagonally.

④ $x(x - 3) = x^2 - 3x$

Multiply
what
is
left.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Multiply	Ko'paytirish	Умножение
Divide	Bo'lish	Деление
Product	Ko'paytma	Произведение
Quotient	Bo'linma	Частное
Reciprocal fraction	Teskari kasr	Обратная дробь
Power of a fraction	Kasrning darajasi	Степень дроби
Cancel across	Krest bo'ylab qisqartirish	Сократить накрест
Divisor	Bo'luvchi	Делитель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

a b : c d equals:

$$\frac{ac}{bd}$$

$$\frac{ad}{bc}$$

$$\frac{bc}{ad}$$

$$\frac{a + d}{b + c}$$

x 2 : x 4 equals:

$$\frac{1}{2}$$

$$2$$

$$\frac{x^2}{8}$$

$$8$$

$(\underline{3x} \ 2y)^2$ equals:

$$\frac{3x^2}{2y^2}$$

$$\frac{9x^2}{4y^2}$$

$$\frac{6x^2}{4y^2}$$

$$\frac{9x}{4y}$$

Before multiplying two algebraic fractions you should first:

multiply out both numerators

find a common denominator

factorise everything and cancel

add the denominators

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{2}{x} \cdot \frac{x}{3}$

ANSWER $\frac{2}{3}$

2. $\frac{a}{b} \cdot \frac{b}{a}$

ANSWER 1

3. $\frac{3}{x} \cdot \frac{x^2}{6}$

ANSWER $\frac{x}{2}$

4. $\frac{x}{2} : \frac{x}{4}$

ANSWER 2

5. $\frac{a}{3} : \frac{a}{6}$

ANSWER 2

6. $\frac{2x}{y} \cdot \frac{y}{4}$

ANSWER $\frac{x}{2}$

7. $\frac{5}{a} : \frac{10}{a^2}$

ANSWER $\frac{a}{2}$

1. $\frac{x+1}{x} \cdot \frac{x}{x+1}$

ANSWER 1

2. $\frac{x^2-1}{x} \cdot \frac{x}{x-1}$

ANSWER $x+1$

3. $\frac{a^2}{b} : \frac{a}{b^2}$

ANSWER ab

4. $\frac{x-2}{3} \cdot \frac{6}{x-2}$

ANSWER 2

5. $\frac{x^2-9}{x} \cdot \frac{x^2}{x+3}$

ANSWER $x(x-3)$

6. $\frac{2a}{a+b} : \frac{4a^2}{a^2-b^2}$

ANSWER $\frac{a-b}{2a}$

7. $\frac{x^2+2x}{x-3} : \frac{x+2}{x^2-9}$

ANSWER $x(x+3)$

Hard · stretch and reason

7 PROBLEMS

$$1. \quad \frac{x^2 - 4}{x^2 + 4x + 4} \cdot \frac{x + 2}{x - 2}$$

ANSWER 1, $x \neq \pm 2$

$$2. \quad \frac{a^3 - b^3}{a^2 - b^2} : \frac{a^2 + ab + b^2}{a + b}$$

ANSWER 1, $a \neq \pm b$

$$3. \quad \frac{x^2 - 5x + 6}{x^2 - 4} \cdot \frac{x^2 + 2x}{x - 3}$$

ANSWER X

$$4. \quad \frac{2x^2 + x - 1}{x^2 - 1} \cdot \frac{x + 1}{2x - 1}$$

ANSWER $\frac{x + 1}{x - 1}$

$$5. \quad \left(1 - \frac{1}{x}\right) : \frac{x - 1}{x^2}$$

ANSWER X

$$6. \quad \frac{m^2 - n^2}{m^2 + mn} : \frac{m - n}{m}$$

ANSWER 1, $m \neq 0, m \neq \pm n$

$$7. \quad \frac{x^2 - 9}{x^2 - 6x + 9} : \frac{x + 3}{x - 3}$$

ANSWER 1, $x \neq 3, x \neq -3$

07 Homework

Homework — 6 problems

Algebra 8, §5, pp. 27–29. Factorise before multiplying, every time.

1. $\frac{4}{x} \cdot \frac{x^2}{8}$

2. $\frac{a}{5} : \frac{a^2}{10}$

3. $\frac{x^2 - 25}{x} \cdot \frac{x}{x - 5}$

4. $\frac{x + 4}{x^2 - 16} \cdot (x - 4)$

5. $\frac{3a}{a - b} : \frac{9a^2}{a^2 - b^2}$

6. $\left(\frac{2x}{3y} \right)^3$

Identical transformation of fractional-rational expressions

Everything from §§2–5 in one question: the long chains where order of operations decides whether the work takes three lines or thirteen.

LESSON 20–22

3 × 40 MIN

ALGEBRA 8, §6

STAGE 9 · 2.3, 2.6

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Simplify a fractional-rational expression that mixes all four operations.
- Deal with a compound (“stacked”) fraction.
- Choose an order of work that keeps the expressions small.
- Prove a given identity by transforming one side into the other.

TEXTBOOK

National	§6, pp. 30–33
Cambridge	Learner’s Book pp. 30–39, 45–53
Workbook	Workbook 2.6

01

Explanation

What “identical transformation” means

DEFINITION

Two expressions are **identically equal** if they take the same value for every permissible value of the variables. Replacing an expression by an identically equal one is an **identical transformation**.

Cancelling, building up to a common denominator, adding, multiplying — each of these is an identical transformation. That is exactly why the permissible values must be stated: outside them the two expressions are *not* the same, because one of them does not exist.

Order of operations

The order is the ordinary one — brackets, then multiplication and division, then addition and subtraction — but with fractions it pays to look ahead:

1. **Tidy every bracket first** into a single fraction.

2. Do the multiplications and divisions, cancelling as you go.
3. Do the additions and subtractions last, over the LCD.
4. Cancel the final answer. Check the permissible values against the *original*.

THE LONG WAY ROUND

Multiplying everything out at the start turns a two-line problem into a page of algebra. Almost every expression in this section is designed to collapse — if your working is growing, you have taken the wrong route.

Compound fractions

A fraction whose numerator or denominator is itself a fraction is a **compound fraction**. The main fraction bar means “divide”:

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x} + \frac{1}{y}} = \left(\frac{1}{x} - \frac{1}{y} \right) : \left(\frac{1}{x} + \frac{1}{y} \right)$$

Combine the top into one fraction, combine the bottom into one fraction, then divide:

$$\frac{y-x}{xy} : \frac{y+x}{xy} = \frac{y-x}{xy} \cdot \frac{xy}{y+x} = \frac{y-x}{y+x}$$

Proving an identity

To prove $A = B$, transform the **more complicated** side until it becomes the simpler one. Never move terms across the equals sign as if it were an equation — you are not solving, you are simplifying.

02 Worked examples

EXAMPLE 1 Simplify $\frac{a+b}{a-b} - \frac{a-b}{a+b}$

① $LCD = (a-b)(a+b) = a^2 - b^2$
Different factors, so multiply them.

②
$$\frac{(a+b)^2 - (a-b)^2}{a^2 - b^2}$$

Build both up. Keep the bracket on the subtracted square.

③ $(a+b)^2 - (a-b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$
The squares cancel; only the middle terms survive.

④
$$\frac{4ab}{a^2 - b^2}$$

Nothing left to cancel.

ANSWER

$$\frac{4ab}{a^2 - b^2}, \quad a \neq \pm b$$

Simplify**EXAMPLE 2**

$$\frac{1 - \frac{1}{x}}{1 - \frac{1}{x^2}}$$

$$\textcircled{1} \quad 1 - \frac{1}{x} = \frac{x-1}{x}$$

Top, as one fraction.

$$\textcircled{2} \quad 1 - \frac{1}{x^2} = \frac{x^2-1}{x^2}$$

Bottom, as one fraction.

$$\textcircled{3} \quad \frac{x-1}{x} \div \frac{x^2-1}{x^2} = \frac{x-1}{x} \cdot \frac{x^2}{(x-1)(x+1)}$$

Divide, and factorise $x^2 - 1$.

$$\textcircled{4} \quad \text{cancel } (x-1) \text{ and one } x$$

$$\textcircled{5} \quad \frac{x}{x+1}$$

ANSWER

$$\frac{x}{x+1}, \quad x \neq 0, x \neq \pm 1$$

EXAMPLE 3 *Prove that $(x + \frac{1}{x})^2 - (x - \frac{1}{x})^2 = 4$*

- ① Use $A^2 - B^2 = (A - B)(A + B)$ with $A = x + \frac{1}{x}$ and $B = x - \frac{1}{x}$.

Far quicker than expanding both squares.

- ② $A - B = \frac{2}{x}$

The x terms cancel.

- ③ $A + B = 2x$

The $1/x$ terms cancel.

- ④ $\frac{2}{x} \cdot 2x = 4$

The x cancels — a constant, for every permissible x.

ANSWER

4, $x \neq 0$

03 Interactive model

These four are the pattern the control work will use. Walk the class through the choice of route, not just the algebra.

Long chains, one step at a time

$$\left(\frac{1}{x} - \frac{1}{y} \right) : \left(\frac{1}{x} + \frac{1}{y} \right)$$

$$\textcircled{1} \quad \frac{1}{x} - \frac{1}{y} = \frac{y - x}{xy}$$

Combine
the
first
bracket.

$$\textcircled{2} \quad \frac{1}{x} + \frac{1}{y} = \frac{y + x}{xy}$$

Combine
the
second.

$$\textcircled{3} \quad \frac{y - x}{xy} \cdot \frac{xy}{y + x}$$

Division
becomes
multiplication
by
the
reciprocal.

$$\textcircled{4} \quad \frac{y - x}{y + x}$$

The
whole
of
xy
cancels.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Rational expression	Ratsional ifoda	Рациональное выражение
Fractional-rational expression	Kasr-ratsional ifoda	Дробно-рациональное выражение
Compound fraction	Murakkab kasr	Многоэтажная дробь
Identity	Ayniyat	Тождество
Identically equal	Ayniy teng	Тождественно равны
Prove	Isbotlash	Доказать
Order of operations	Amallar tartibi	Порядок действий
Transform	Almashtirish	Преобразовать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

$\frac{a}{a-b} + \frac{b}{b-a}$ equals:

0

1

$$\frac{a+b}{a-b}$$

-1

To simplify a compound fraction you should:

cancel the small fractions first

combine the top and the bottom, then divide

multiply everything out

add the numerators

$(x + \frac{1}{x})^2 - (x - \frac{1}{x})^2$ equals:

0

2

4

 $4x^2$

Proving an identity means:

solving for x

transforming one side into the other

moving terms across the equals sign

substituting one value of x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

$$1. \quad \left(\frac{1}{x} + \frac{1}{y} \right) \cdot xy$$

ANSWER $y + x$

$$2. \quad 1 + \frac{1}{x}$$

ANSWER $\frac{x+1}{x}$

$$3. \quad \frac{a}{b} + 1$$

ANSWER $\frac{a+b}{b}$

$$4. \quad \frac{x}{2} + \frac{x}{3}$$

ANSWER $\frac{5x}{6}$

$$5. \quad 1 - \frac{1}{2x}$$

ANSWER $\frac{2x-1}{2x}$

$$6. \quad \frac{x+1}{2} \cdot \frac{4}{x+1}$$

ANSWER 2

$$7. \quad \frac{2}{x} : \frac{4}{x^2}$$

ANSWER $\frac{x}{2}$

Medium · the standard question

7 PROBLEMS

$$1. \quad \left(\frac{1}{x} - \frac{1}{y} \right) : \left(\frac{1}{x} + \frac{1}{y} \right)$$

ANSWER $\frac{y-x}{y+x}$

$$2. \quad \left(1 + \frac{1}{x} \right) \cdot \frac{x}{x+1}$$

ANSWER 1

$$3. \quad x - \frac{1}{x}$$

ANSWER $\frac{x^2-1}{x}$

$$4. \quad \frac{a+b}{a-b} - \frac{a-b}{a+b}$$

ANSWER $\frac{4ab}{a^2-b^2}$

$$5. \quad \left(\frac{1}{x-1} + \frac{1}{x+1} \right) \cdot \frac{x^2-1}{2}$$

ANSWER x

$$6. \quad \frac{x^2-1}{x} : (x+1)$$

ANSWER $\frac{x-1}{x}$

$$7. \quad \left(\frac{1}{a} + \frac{1}{b} \right) : \frac{1}{ab}$$

ANSWER a + b

Hard · stretch and reason

7 PROBLEMS

$$1. \quad \left(\frac{1}{x-1} - \frac{1}{x+1} \right) : \frac{2}{x^2-1}$$

ANSWER 1, $x \neq \pm 1$

$$2. \quad \left(x + \frac{1}{x} \right)^2 - \left(x - \frac{1}{x} \right)^2$$

ANSWER 4

$$3. \quad \frac{a}{a-b} + \frac{b}{b-a}$$

ANSWER 1, $a \neq b$

$$4. \quad \frac{1 - \frac{1}{x}}{1 - \frac{1}{x^2}}$$

ANSWER $\frac{x}{x+1}$

$$5. \quad \left(\frac{1}{x} + \frac{1}{y} \right) \cdot \frac{xy}{x+y}$$

ANSWER 1

$$6. \quad \text{Prove that } \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$$

ANSWER LCD $n(n+1)$: the numerator is $(n+1) - n = 1$.

$$7. \quad \frac{1}{x^2-x} + \frac{1}{x^2-1}$$

ANSWER $\frac{2x+1}{x(x^2-1)}$

07 Homework

Homework — 5 problems

Algebra 8, §6, pp. 30–33. Choose your route before you start writing.

$$1. \left(\frac{1}{x} - \frac{1}{x+1} \right) \cdot x(x+1)$$

$$2. \frac{x+y}{x-y} + \frac{x-y}{x+y}$$

$$3. \frac{1 + \frac{1}{x}}{1 - \frac{1}{x}}$$

$$4. \left(\frac{a}{b} - \frac{b}{a} \right) : \frac{a^2 - b^2}{ab}$$

$$5. \text{ Prove that } \frac{1}{x-1} - \frac{1}{x+1} = \frac{2}{x^2-1}$$

Control work 2 · Rational expressions

Multiplication, division and long chains, then the work-on-mistakes lesson that turns the paper into teaching.

LESSON 23–24

2 × 40 MIN

ALGEBRA 8, §§5–6

STAGE 9 · UNITS 2, 8

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess multiplication, division, powers and combined transformations.
- Identify which of the four standard errors each lost mark belongs to.
- Re-solve every lost task in full.

TEXTBOOK

National	Revision of §§5–6
Cambridge	Learner’s Book pp. 21–54, 161–189
Workbook	Workbook Units 2, 8

01 Explanation

Lesson 23 — the paper (40 minutes)

Two variants of seven tasks. The balance of the paper:

- tasks 1–2 · multiplication and division of fractions (§5)
- task 3 · a power of a fraction (§5)
- tasks 4–5 · a combined expression with brackets (§6)
- task 6 · a compound fraction (§6)
- task 7 · prove an identity (§6)

Marking: 2 marks per task, 14 in total. Task 7 earns nothing for substituting a single value — an identity has to be proved for all permissible values.

Lesson 24 — work on mistakes (40 minutes)

The errors in this paper fall into four groups, and they are not the same four as in Control work 1:

WHAT GOES WRONG IN §§5–6

1. Not turning the divisor over — multiplying by $\frac{C}{D}$ instead of $\frac{D}{C}$.

2. **Expanding before cancelling**, producing a page of algebra and then an arithmetic slip.
3. **The index not reaching the coefficient** — $(\frac{3x}{2})^2$ written as $\frac{3x^2}{2}$.
4. **Treating an identity as an equation** and moving terms across the equals sign.

The Hard set below is seven pieces of wrong working drawn from these four. Diagnose, then repair.

02 Worked examples

EXAMPLE 1 Find and repair the error: $\frac{x}{3} : \frac{x}{6} = \frac{x^2}{18}$

- ① The divisor was multiplied, not inverted.
This is error 1.

② $\frac{x}{3} \cdot \frac{6}{x}$

Turn the second fraction over.

③ $\frac{6}{3} = 2$

The x cancels.

ANSWER

2, $x \neq 0$

EXAMPLE 2**Find and repair the error:** $\left(\frac{3x}{2}\right)^2 = \frac{3x^2}{2}$

- ① The index must reach every factor, above and below.
This is error 3.

- ② $\left(\frac{3x}{2}\right)^2 = \frac{(3x)^2}{2^2}$
Square the whole numerator and the whole denominator.

- ③ $\frac{9x^2}{4}$
 $3^2 = 9$ and $2^2 = 4$.

ANSWER

$$\frac{9x^2}{4}$$

03 Interactive model

Lesson 24: show a piece of wrong working, take a vote on which of the four errors it is, then reveal.

Diagnose the error

Claimed: $\frac{a}{b} : \frac{c}{d} = \frac{ac}{bd}$

- ① Dividing by a fraction means multiplying by its reciprocal.

② $\frac{a}{b} \cdot \frac{d}{c}$

The
second
fraction
turns
over.

③ $\frac{ad}{bc}$

Compare
with
the
claim
—
the
c
and
d
are
the
wrong
way
round.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Reciprocal	Teskari	Обратный
Coefficient	Koeffitsiyent	Коэффициент
Expand	Ochish (yoyish)	Раскрыть
Diagnose	Aniqlash	Определить (диагностировать)
Justify	Asoslash	Обосновать
Identity	Ayniyat	Тождество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\frac{x}{3} : \frac{x}{6}$ equals:

$$\frac{x^2}{18}$$

$$2$$

$$\frac{1}{2}$$

$$\frac{x}{18}$$

$(2a - 5)^3$ equals:

$$\frac{8a^3}{125}$$

$$\frac{2a^3}{125}$$

$$\frac{6a^3}{15}$$

$$\frac{8a^3}{5}$$

The fastest first move on $\frac{x^2 - 4}{x + 2} \cdot \frac{x + 2}{x - 2}$ is:

multiply out the numerators

factorise and cancel

find a common denominator

substitute a value of x

To prove an identity you may:

multiply both sides by the LCD

move a term across the equals sign

transform one side into the other

test one value of x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** $\frac{3}{x} \cdot \frac{x^2}{9}$

ANSWER $\frac{x}{3}$

2. **Task 2.** $\frac{a}{4} : \frac{a}{8}$

ANSWER 2

3. **Task 3.** $\left(\frac{2x}{3}\right)^2$

ANSWER $\frac{4x^2}{9}$

4. **Task 4.** $\frac{x^2 - 1}{x} \cdot \frac{x}{x + 1}$

ANSWER $x - 1$

5. **Task 5.** $1 + \frac{2}{x}$

ANSWER $\frac{x + 2}{x}$

6. **Task 6.** $1 + \frac{\frac{1}{x}}{x + 1}$

ANSWER $\frac{1}{x}, x \neq 0, -1$

7. **Task 7. Prove** $\frac{1}{x} - \frac{1}{x + 1} = \frac{1}{x(x + 1)}$

ANSWER LCD $x(x + 1)$; numerator $(x + 1) - x = 1$.

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** $\frac{5}{a} \cdot \frac{a^2}{10}$

ANSWER $\frac{a}{2}$

2. **Task 2.** $\frac{x}{6} : \frac{x}{3}$

ANSWER $\frac{1}{2}$

3. **Task 3.** $\left(\frac{3a}{4b}\right)^2$

ANSWER $\frac{9a^2}{16b^2}$

4. **Task 4.** $\frac{x^2 - 9}{x + 3} \cdot \frac{x}{x - 3}$

ANSWER $x, x \neq \pm 3$

5. **Task 5.** $\frac{a}{a + b} + \frac{b}{a + b}$

ANSWER $1, a \neq -b$

6. **Task 6.**
$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x} + \frac{1}{y}}$$

ANSWER $\frac{y - x}{y + x}$

7. **Task 7. Prove** $\frac{a}{a-b} + \frac{b}{b-a} = 1$

ANSWER $b - a = -(a - b)$, so the sum is $\frac{a-b}{a-b} = 1$.

Hard · stretch and reason

7 PROBLEMS

1. Find the error: $\frac{a}{b} : \frac{c}{d} = \frac{ac}{bd}$

ANSWER The divisor was not inverted. Correct: $\frac{ad}{bc}$.

2. Find the error: $(\frac{3x}{2})^2 = \frac{3x^2}{2}$

ANSWER The index missed the 3 and the 2. Correct: $\frac{9x^2}{4}$.

3. Find the error: $\frac{x^2 - 4}{x + 2} \cdot \frac{x + 2}{x - 2} = x^2 - 4$

ANSWER Nothing was cancelled. Correct: 1, for $x \neq \pm 2$.

4. Find the error: to prove $\frac{1}{x - 1} - \frac{1}{x + 1} = \frac{2}{x^2 - 1}$, “multiply both sides by $x^2 - 1$ ”

ANSWER An identity is proved by transforming one side, not by operating on both.

5. Find the error: $\frac{x}{2} : 2 = x$

ANSWER Dividing by 2 means multiplying by $\frac{1}{2}$. Correct: $\frac{x}{4}$.

6. Find the error: $(\frac{1}{x} + \frac{1}{y})^{-1} = x + y$

ANSWER The reciprocal of a sum is not the sum of reciprocals. Correct: $\frac{xy}{x + y}$.

7. Find the error: $\frac{x^2}{x} : x = x$

ANSWER $\frac{x^2}{x} = x$, and $x : x = 1$. Correct answer 1, for $x \neq 0$.

07 Homework

Homework after the work-on-mistakes lesson

Re-solve every task you lost marks on, then these three.

1. $\frac{x^2 - 16}{x} : \frac{x + 4}{x^2}$

2. $(\frac{1}{a} - \frac{1}{b}) : \frac{b - a}{ab}$

3. Write out the four errors listed in this lesson with one example of each, in your own words.

The function $y = k / x$ — properties and graph

Inverse proportion: the first function of the course whose graph is not a straight line, and whose defining feature is a product that never changes.

LESSON 25–27

3 × 40 MIN

ALGEBRA 8, §7

STAGE 9 · 10.1–10.2, 11.2

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

LEARNING OBJECTIVES

- Recognise inverse proportion and find k from one point.
- Draw the graph of $y = k/x$ by table of values.
- State the properties: domain, the two branches, quadrants, symmetry, asymptotes.
- Solve practical problems in which a product is constant.

TEXTBOOK

National §7, pp. 34–38

Cambridge Learner's Book
pp. 213–224, 242–249Workbook Workbook 10.1,
11.2

01 Explanation

Inverse proportion

DEFINITION

Two quantities are **inversely proportional** if their product is constant. Writing that constant as k , we get $xy = k$, that is

$$y = \frac{k}{x}, \quad k \neq 0, x \neq 0$$

Everyday examples: a fixed distance covered at different speeds ($t = \frac{s}{v}$); a fixed

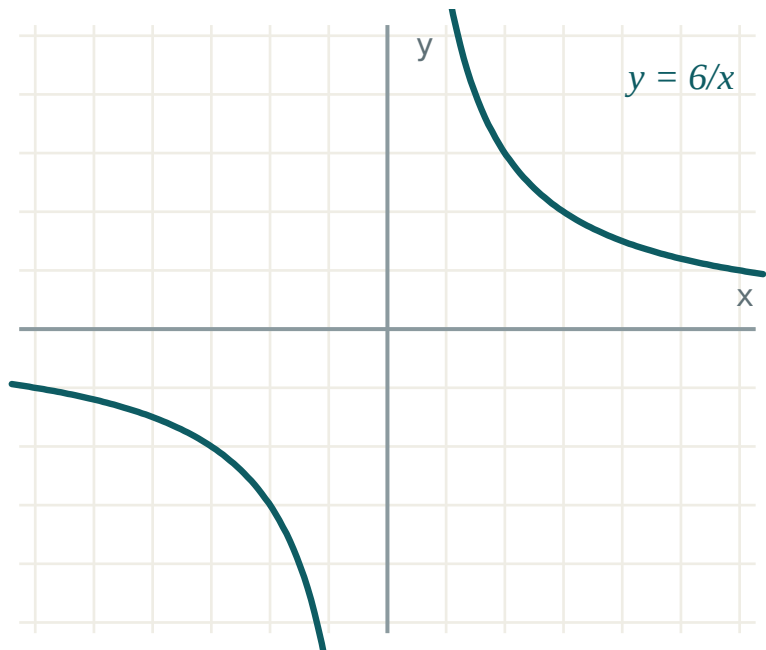
amount of work shared between different numbers of workers; a rectangle of fixed area ($h = \frac{s}{b}$).

If x doubles, y halves. If x is multiplied by 3, y is divided by 3. The product stays put.

The graph — a hyperbola

Take $k = 6$ and build a table:

x	1	2	3	6	-1	-2	-3	-6
y	6	3	2	1	-6	-3	-2	-1



The graph of $y = 6/x$. Two separate branches — never one connected curve.

The curve is called a **hyperbola**, and each half is a **branch**.

The properties

- **Domain:** every x except 0 . The value $x = 0$ is not merely awkward — there is no point of the graph above or below the origin at all.
- **Range:** every y except 0 . The curve never touches either axis.
- **Quadrants:** if $k > 0$ the branches lie in quadrants **I and III**; if $k < 0$, in **II and IV**.
- **Behaviour:** for $k > 0$ the function is decreasing on each branch separately — but *not* on the whole domain, because the two branches are not joined.
- **Asymptotes:** the curve comes arbitrarily close to both axes and never meets them.
- **Symmetry:** the graph is symmetric about the origin, and about the line $y = x$ (or $y = -x$ when $k < 0$).

SAY THIS OUT LOUD IN CLASS

“Decreasing on each branch” is not the same as “decreasing”. Take $y = \frac{6}{x}$: at $x = -1$, $y = -6$; at $x = 1$, $y = 6$. The value went *up* as x went up — because we crossed the gap at $x = 0$.

Finding k

One point is enough. If the graph passes through $(3, 4)$ then $k = 3 \cdot 4 = 12$, so $y =$

$$\frac{12}{x}$$

And the test for whether a point lies on the graph is just as quick: multiply its coordinates and compare with k .

02 Worked examples

EXAMPLE 1 The graph of $y = k/x$ passes through $(-2, 5)$. Find k and describe the graph.

① $k = xy = (-2)(5) = -10$

The product of the coordinates is k .

② $y = -\frac{10}{x}$

The rule.

③ $k < 0$

so the branches lie in quadrants II and IV.

④ Domain $x \neq 0$, range $y \neq 0$.

The axes are asymptotes.

ANSWER

$k = -10$; a hyperbola in quadrants II and IV.

EXAMPLE 2 Find where $y = 8/x$ meets $y = 2x$

$$\textcircled{1} \quad \frac{8}{x} = 2x$$

Set the two rules equal.

$$\textcircled{2} \quad 8 = 2x^2$$

Multiply both sides by x — legal, since $x \neq 0$.

$$\textcircled{3} \quad x^2 = 4, \text{ so } x = 2 \text{ or } x = -2$$

Both roots are permissible.

$$\textcircled{4} \quad x = 2 \implies y = 4; x = -2 \implies y = -4$$

Substitute back.

ANSWER

$(2, 4)$ and $(-2, -4)$

EXAMPLE 3 A rectangle has area 36 cm^2 . Express its height in terms of its base and name the graph.

① $b \cdot h = 36$

Area of a rectangle.

② $h = \frac{36}{b}$

Make h the subject.

③ $k = 36 > 0$, and lengths are positive.

So only the branch in quadrant I has meaning here.

④ A hyperbola — but only the part with $b > 0$ is drawn.

Context can cut a graph down.

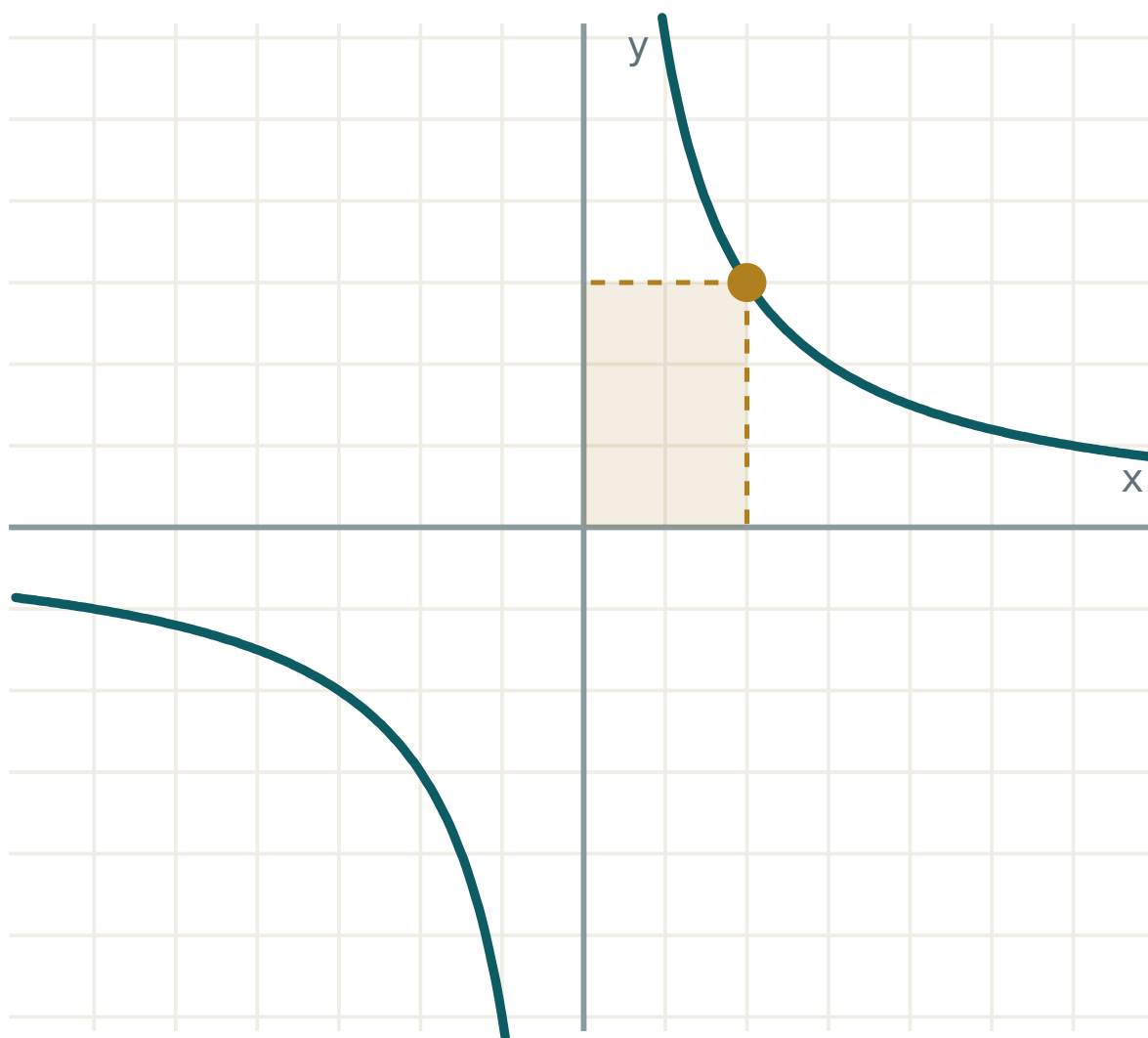
ANSWER

$h = \frac{36}{b}$; one branch of a hyperbola, $b > 0$

03 Interactive model

Drag k through zero and ask the class to predict which quadrants the branches jump to.

The graph of $y = k/x$



k 6

x 2

 $y = k/x$ 3 $x \cdot y$ 6

quadrants I and III

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Function	Funksiya	Функция
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Coefficient k	k koeffitsiyenti	Коэффициент k
Graph	Grafik	График
Hyperbola	Giperbola	Гипербола
Branch	Tarmoq	Ветвь
Domain	Aniqlanish sohasi	Область определения
Asymptote	Asimptota	Асимптота
Quadrant	Koordinata choragi	Координатная четверть
Symmetry	Simmetriya	Симметрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The graph of $y = \underline{7} x$ lies in quadrants:

I and II

I and III

II and IV

all four

If $y = \underline{k} x$ passes through $(3, 4)$ then k is:

12

$\frac{3}{4}$

7

$\frac{4}{3}$

For $y = \underline{k} x$, if x is multiplied by 4 then y is:

multiplied by 4

divided by 4

unchanged

multiplied by 16

The graph of $y = \frac{k}{x}$ meets the x-axis:

at one point

at two points

at the origin

never

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $y = \frac{12}{x}$. Find y when $x = 4$

ANSWER $y = 3$

2. $y = \frac{6}{x}$. Find y when $x = -2$

ANSWER $y = -3$

3. Does the point $(2, 5)$ lie on $y = \frac{10}{x}$?

ANSWER Yes — $2 \cdot 5 = 10$.

4. In which quadrants does $y = \frac{7}{x}$ lie?

ANSWER I and III

5. In which quadrants does $y = -\frac{5}{x}$ lie?

ANSWER II and IV

6. $y = \frac{8}{x}$. Find x when $y = 2$

ANSWER $x = 4$

7. Is $y = \frac{3}{x}$ defined at $x = 0$?

ANSWER No — $x = 0$ is not in the domain.

Medium · the standard question

7 PROBLEMS

1. $(3, 4)$ lies on $y = k/x$. Find k .

ANSWER $k = 12$

2. $(-2, 5)$ lies on $y = k/x$. Find k .

ANSWER $k = -10$

3. For $y = k/x$, what happens to y when x doubles?

ANSWER y halves — the product xy is constant.

4. Find k so that $y = k/x$ passes through $(0.5, 8)$

ANSWER $k = 4$

5. Complete the table for $y = \frac{12}{x}$ at $x = 1, 2, 3, 4, 6, 12$

ANSWER 12, 6, 4, 3, 2, 1

6. $(2, 9)$ and $(a, 6)$ both lie on $y = k/x$. Find a .

ANSWER $k = 18$, so $a = 3$

7. For $y = \frac{24}{x}$, list the positive integers x giving an integer y

ANSWER 1, 2, 3, 4, 6, 8, 12, 24

Hard · stretch and reason

7 PROBLEMS

1. Show that for $y = k/x$ the product xy is the same at every point

ANSWER $xy = x \cdot \frac{k}{x} = k$ for every permissible x .

2. $y = k/x$ passes through $(-3, -4)$. Find k and the quadrants.

ANSWER $k = 12$; quadrants I and III.

3. A rectangle has area 36. Write h in terms of b and name the graph.

ANSWER $h = \frac{36}{b}$; a hyperbola, $b > 0$ only.

4. (m, n) lies on $y = k/x$. Show that (n, m) does too.

ANSWER $nm = mn = k$, so the second point satisfies the same rule.

5. Solve $\frac{6}{x} = x$

ANSWER $x^2 = 6$, so $x = \pm\sqrt{6}$

6. Find where $y = \frac{8}{x}$ meets $y = 2x$

ANSWER $(2, 4)$ and $(-2, -4)$

7. Sketch $y = \frac{4}{x}$ and $y = -\frac{4}{x}$ together and describe the symmetry

ANSWER Each is the reflection of the other in either coordinate axis.

07 Homework

Homework — 6 problems

Algebra 8, §7, pp. 34–38. Draw the graph of question 4 on squared paper.

1. $y = \frac{15}{x}$. Find y when $x = 3$, $x = 5$ and $x = -15$
2. $(4, 6)$ lies on $y = k/x$. Find k and write the rule.
3. In which quadrants does $y = -\frac{9}{x}$ lie? Explain in one sentence.
4. Draw the graph of $y = \frac{4}{x}$ for x from -4 to 4 , using a table of values.
5. A journey of 120 km takes t hours at speed v km/h. Write t in terms of v and find t when $v = 60$.
6. Solve $\frac{10}{x} = \frac{x}{10}$